PAX4 — "retina development in camera-type eye" (GO:0060041): Focused Curation Review

Gene: PAX4 (human), UniProt O43316 **Hypothesis evaluated:** *PAX4 has retina development in camera-type eye (GO:0060041).* **Focus type:** function_assignment **Existing annotation under review:** IBA (ECO:0000318), GO_REF:0000033 (PANTHER/PAINT ancestral node)

Executive Judgment

Verdict: Over-annotated / weakly supported — the term should NOT be retained as a confident PAX4 function.

The GO:0060041 annotation on PAX4 is **IBA-only**, inferred phylogenetically from a group-III PAX ancestral node whose retina-development signal is driven by the paralog/ortholog **PAX6** (confirmed: the "with/from" support includes MGI:97490 = mouse *Pax6*, the canonical eye master gene). There is **no direct experimental evidence** that human PAX4 participates in retina/eye development:

- Human **complete PAX4 loss-of-function** produces **neonatal diabetes** (a beta-cell phenotype) with **no reported ocular defect** (P 40614820).
- **Pax4-null mice** show ghrelin cells replacing beta cells — a **pancreatic islet** fate defect, no retina phenotype (P 14970313).
- The **only** PAX4-specific eye study (rat; P 19012751) shows Pax4 is **absent in the fetal eye** and rises **postnatally** in **differentiated**

> **photoreceptors** with a **diurnal rhythm** — i.e., mature-photoreceptor physiology, **not retinal morphogenesis**.

PAX4's core, well-documented function is **pancreatic beta-cell specification/identity**. The retina-development role experimentally belongs to **PAX6**. This is a textbook case of **paralog/family over-annotation by IBA propagation**.

Most important caveats: (1) absence of a reported eye phenotype is not formal proof of no role; (2) a single rat expression study does show *Pax4* in photoreceptors, so a *non-developmental* mature-photoreceptor role cannot be excluded; (3) GO_REF:0000033 IBA is a legitimate, curator-reviewed inference method, so removal should be a PAINT-level decision, not unilateral.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Confi limit
QuickGO record for O43316 (GO_REF:0000033) + NCBI E-utilities resolution	Database / evolutionary	**Qualifies → weakens**	Is the annotation direct or inferred?	GO:0060041 is **IBA**; "with" = MGI:97490 (**mouse Pax6**), PTN002900580, zebrafish pax orthologs; PAX4's own IDA/ TAS annotations have **no** retina term	Database; programmatic	High prove show paral
P 40614820	Mutant phenotype (human)	**Refutes (core role)**	Does PAX4 loss affect the eye?	Complete human PAX4 LoF → transient **neonatal diabetes**; role in **beta-cell** development; no ocular defect reported	Human patients; iPSC/ EndoC-βH1	High; of-ph cavea
P 14970313	Mutant phenotype (mouse)	**Refutes (core role)**	What does Pax4 loss cause?	Pax4-mutant mice: **ghrelin cells replace beta cells** (islet fate), no retina phenotype	Mouse pancreas	High panc retin exam
P 19012751 (Rath 2009)	Localization/ expression (rat)	**Qualifies / competing**	Is Pax4 in the retina, and when?	Pax4 **absent in fetal eye**, rises **postnatally**, exclusive to **photoreceptors**, **diurnal** rhythm	Rat retina	Mode only; expre devel funct suppo *matu*

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						not GO:0
P 40558032 (review)	Review/ database	**Qualifies (core role)**	What is PAX4's principal function?	PAX4 expressed **mainly in pancreas**; key regulator of **beta-cell** differentiation	Human	Revie orien
P 42210309; P 39118533	Direct assay / mutant	Supports core (pancreatic) role	PAX4 function	R192H impairs beta-cell identity; PAX4 delivery improves T2D islet function	Mouse/ human islets	High panc funct
P 19146846 (mouse *Pax6*, IGI); P 8951055 (mouse *Pax2*, IMP); P 20152834 (zebrafish *pax6a*, IMP)	Mutant phenotype (orthologs)	**Competing / explains**	Where does the ancestral GO:0060041 signal come from?	Experimental GO:0060041 is anchored in **Pax6/Pax2** eye genes; human PAX4 **and** human PAX6 carry GO:0060041 **only by IBA**	Mouse/ zebrafish eye	High; pinpo paral of the PAX4 exper retin anno

GO Curation Implications

Lead (requires curator verification): GO:0060041 "retina development in camera-type eye" on PAX4 is a **BP** term supported **only by IBA** whose evidence base is the paralog PAX6. Recommended action options, in order of preference:

1. **Do not retain / flag for removal at the PAINT level.** The ancestral-node annotation appears to over-reach to the PAX4 branch. If the PAINT curator agrees PAX4 lacks direct eye function, the leaf IBA should be removed or the ancestral annotation restricted to the PAX6/PAX2-containing subtree.
2. **If retained, treat as NON-CORE and low-confidence**, explicitly annotated as family-inferred, and do **not** elevate to an experimental evidence code.
3. **Do NOT create an experimental (IDA/IMP) retina-development annotation** — no such evidence exists.

The **core** PAX4 annotations that *are* experimentally grounded are pancreatic/beta-cell BP terms (e.g., GO:0003309 type B pancreatic cell differentiation) — these should be prioritized as the gene's primary function.

If any eye-related annotation were ever justified for PAX4, the rat data would point toward a *mature photoreceptor* / circadian-physiology role, **not** GO:0060041 (a developmental term) — but that evidence is rat-only and insufficient for a human annotation.

Mechanistic Scope

- **Direct molecular function:** PAX4 is a paired-box + homeodomain **transcriptional repressor/regulator** that specifies and maintains **pancreatic beta- (and delta-) cell** identity, counteracting ARX (alpha fate) and interacting with PAX6/PDX1/NKX2.2 networks.
- **Tested here (GO:0060041):** a **developmental BP** — participation in camera-type-eye retina morphogenesis. This would require PAX4 to act during retinal development. Evidence shows the opposite temporally (absent in fetal eye).
- **Downstream vs direct:** No PAX4-driven retinal developmental program is documented. The rat photoreceptor expression is **post-developmental** and likely reflects a distinct, mature-tissue transcriptional role — not the developmental process named by GO:0060041.

Conflicts and Alternatives

- **Paralog confusion (primary alternative, now pinpointed):** The experimental GO:0060041 annotations in this gene family belong to **mouse *Pax6*** (IGI, P 19146846), **mouse *Pax2*** (IMP, P 8951055) and **zebrafish *pax6a*** (IMP, P 20152834) — all canonical eye genes. Both human PAX4 **and** human PAX6 carry GO:0060041 **only by IBA** (GO_REF:0000033). The IBA "with" field for PAX4 literally cites mouse *Pax6* (MGI:97490). The retina signal is therefore Pax6/Pax2's, propagated across the PAX node to PAX4, which has **no** experimental retina annotation.
- **Organism specificity:** The only positive eye signal is in **rat**; not replicated in human. Human/mouse PAX4 LoF phenotypes are pancreatic.
- **Expression ≠ function artifact:** Rath 2009 is localization-only; it cannot establish a developmental role and is interpreted by its own authors as mature-photoreceptor/circadian function.
- **Database carry-over:** This is the expected behavior of PAINT/IBA for a gene family where one member (PAX6) has strong eye evidence.

Knowledge Gaps

1. **Is there any PAX4 retinal phenotype?** Checked: no eye phenotype reported in human NDM patients (P 40614820) or Pax4-null mice (P 14970313), but neither study systematically examined the retina. *Resolve with:* targeted

ophthalmologic/histologic phenotyping of Pax4-null retinas and review of human carriers' ocular records.

2. **Does human PAX4 share the rat photoreceptor expression?** Checked: rat-only data. *Resolve with:* human retina scRNA-seq (e.g., retinal cell atlases) / GTEx photoreceptor-relevant tissues to test photoreceptor expression and timing.
3. **If expressed in photoreceptors, what is the function?** Not tested anywhere. *Resolve with:* photoreceptor-specific Pax4 perturbation + ERG/circadian assays — would distinguish a mature-physiology role from GO:0060041.
4. **PAINT ancestral-node scope.** *Resolve with:* inspection of the PANTHER family tree (PTN002900580) to confirm whether GO:0060041 was placed on a node ancestral to both PAX6 and PAX4.

Discriminating Tests

1. **Human retina single-cell atlas query** (e.g., CELLxGENE / Human Retina Cell Atlas): is PAX4 detectably expressed in human photoreceptors, and when? Cheap, directly tests organism generalizability.
2. **Phylo-tree inspection of PANTHER node PTN002900580**: determine if the retina term sits on a node that over-reaches PAX4; the most direct curation-relevant check.
3. **Reanalysis of Pax4-null mouse retina** (histology, markers, ERG): would convert "not examined" into a real negative/positive.
4. **Comparative promoter/target analysis**: does PAX4 regulate canonical retinal-development genes (e.g., CRX, NRL, RHO) in any ChIP/CUT&RUN dataset (cf. EndoC-βH1 CUT&RUN in

   P 40614820

   which found pancreatic targets)?

Curation Leads (require curator verification)

- **Action:** Flag GO:0060041 on PAX4 as **paralog-driven IBA over-annotation**; recommend review at the PANTHER/PAINT ancestral node (PTN002900580) and likely **removal or restriction** of the retina term from the PAX4 branch. Do not retain as a core function.
- **Candidate references to verify:**
- P 19012751 — *exact snippets:* "Pax4 expression is not detectable in the foetal eye; however, postnatal Pax4 transcript levels rapidly increase." and "The daily changes in Pax4 expression may contribute to daily changes in function in the differentiated retinal photoreceptor." (Supports a **mature photoreceptor** role, **not** GO:0060041.)
- P 40614820 — *snippet:* "establishing it as a novel cause of NDM and highlighting its role in human beta cell development." (Core function is pancreatic.)
- P 14970313 — *snippet:* "we demonstrate that Pax4 mutant mice display a similar phenotype" (islet, not eye, phenotype).
- **Candidate alternative terms (core, better supported):** retain/prioritize pancreatic BP terms — e.g., GO:0003309 (type B pancreatic cell differentiation), endocrine pancreas development — over the retina term.
- **Suggested questions for curator:** (1) Does the PAINT node place GO:0060041 above the PAX6/PAX4 split? (2) Is any human photoreceptor expression evidence available to justify even a non-developmental eye annotation? (3) Should a mature-photoreceptor term be considered instead if expression is confirmed?
- **Suggested experiments:** Pax4-null retinal phenotyping + ERG; human retina scRNA-seq expression/timing check.

Provenance note

Annotation provenance was retrieved programmatically from EBI QuickGO (geneProductId O43316) and ortholog identities resolved via NCBI E-utilities (MGI:97490 → Entrez 18508 → mouse *Pax6*). Literature retrieved via PubMed. No result was fabricated; where a resource was not queried (e.g., single-cell atlases, PANTHER tree inspection), it is listed as a gap/discriminating test rather than a result.

Human expression check — attempted, inconclusive (tool limitation). I attempted to quantify human PAX4 tissue expression via the GTEx v8 `medianGeneExpression` API (gencodeId ENSG00000106331.15, resolved through the GTEx reference-gene endpoint). The endpoint returned **zero median records**; a control query returned **zero records for INS and PAX6 as well**, indicating an **API-format limitation of this endpoint, not biological absence**. Therefore no GTEx expression value is reported here (none was fabricated). Independent orientation (not programmatically verified this run): Human Protein Atlas classifies PAX4 as pancreas-enriched with very low/restricted bulk expression (beta-cell restricted), and GTEx bulk tissues do not include neural retina, so GTEx cannot test the rat photoreceptor claim regardless. Confirming human photoreceptor expression therefore remains a knowledge gap best resolved with a human retina single-cell atlas (e.g., CELLxGENE).
